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Structural design, finite element research and site experience from my MSc in Structural Engineering at the University of Leeds and my BEng in Civil Engineering. Projects are grouped by type. Each sheet sets out the brief, my role, the codes and tools I used and what came out of it.

MSc
Structural Engineering, University of Leeds

First Class
BEng (Hons) Civil Engineering

Scopus
Indexed journal paper, July 2025

Patent
FRP emergency shelter system

A · DESIGN PROJECTS

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A1

MSC STRUCTURAL ENGINEERING DESIGN PROJECT · UNIVERSITY OF LEEDS

Vertical Forest, Woodhouse Lane, Leeds

ROLE	LEVEL	CODES	TOOLS
Individual detailed design after a group concept stage	MSc, 2025–26	BS EN 1990 to 1995 and 1997, UK NA	Revit, hand calculations, IStructE carbon calculator

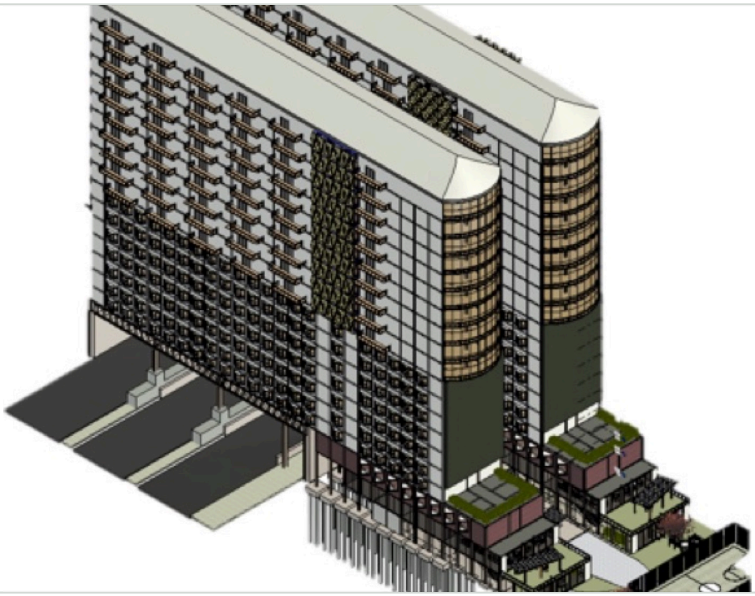
The Woodhouse Lane car park in Leeds sits on a bridge deck over the A58(M) inner ring road. The brief was to replace it with a sustainability-led vertical forest of homes, student rooms and public space, with at least half the building over the existing deck. The precast deck planks cannot take new load, so everything above has to be carried on a new transfer structure.

20
storeys, two stepped modular towers

864
homes and rooms (240 apartments, 624 student rooms)

145
$kgCO_2e/m^2$ embodied carbon (A1–A5)

133
calculation sheets



Chosen scheme: two stepped modular towers over the existing bridge deck

CONCEPT CHOICE

Two concepts went through a weighted decision matrix. A steel frame with hollow core floors scored 270 and needed 16,855 kN on the existing bridge columns, well above their 11,287 kN capacity. Lightweight volumetric modules with timber cassette floors scored 355 and kept the column load to 10,977 kN.

STRUCTURAL SCHEME

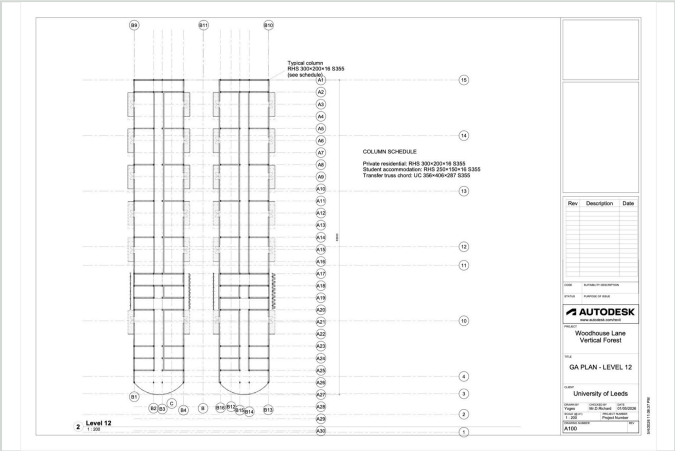
- Off-site modules with RHS corner columns and PFC edge beams in S355.
- C24 timber cassette floors at about a third of the weight of a concrete slab.
- A 2.0 m deep Pratt transfer truss spanning 19.6 m at level 01.
- RC cores with perimeter cross-bracing, Consequence Class 2B.
- 750 mm CFA piles 15 m deep, FRP jacketing and micropile underpinning.

CRITICAL ELEMENT CHECKS (ULS UTILISATION)

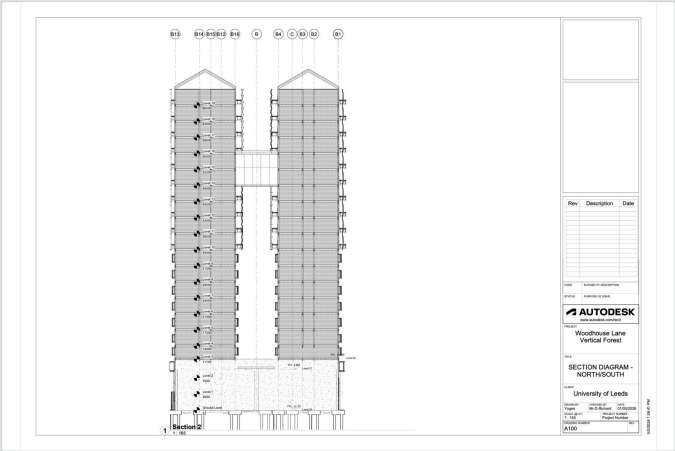
ELEMENT	SECTION	UTILISATION
PFC edge beam, 9.6 m	PFC 380×100×54	0.97
RHS module column	RHS 300×200×16	0.91
Transfer truss chord	UC 356×406×287	0.90
RC core wall	300 mm C32/40	0.84
Timber cassette floor	C24 225×47 @ 400	0.82
CFA pile (GEO)	750 dia × 15 m	0.72
Perimeter bracing	L 100×75×8	0.31

CARBON, DYNAMICS, PROGRAMME AND COST

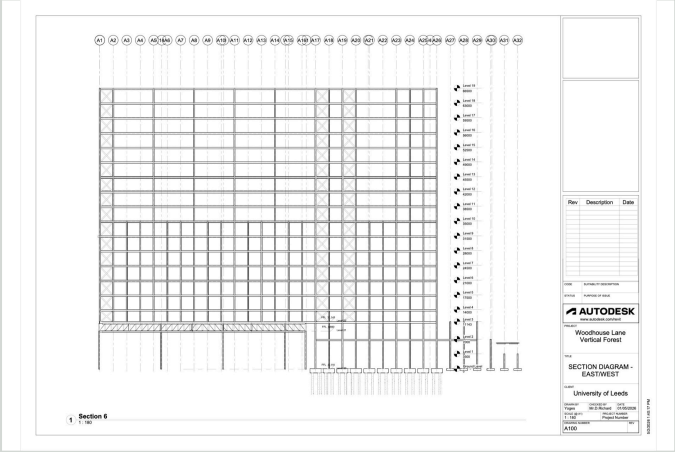
Embodied carbon	≈ 6,100 tCO_2e against 8,200 for the steel frame concept
Benchmark	145 $kgCO_2e/m^2$, inside the LETI 2030 band
Tower frequency	n_1 = 0.70 Hz, floor R = 4.5
Programme	104 weeks, ≈ 20 weeks faster than steel frame
Cost estimate	≈ £125.3m (Spon's 2025)



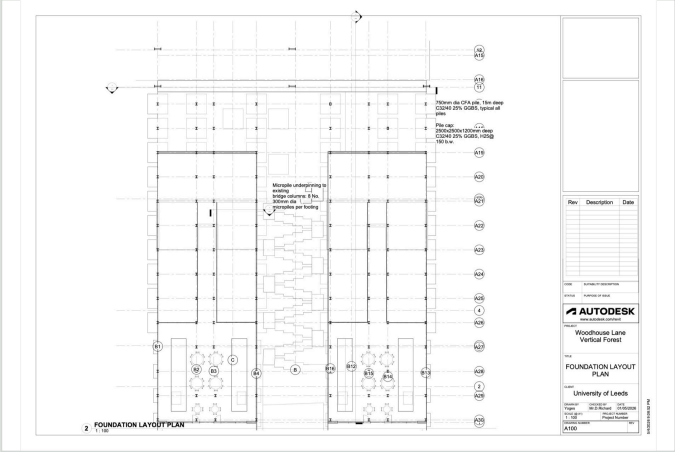
GA plan, Level 12, with column schedule (Revit, drawn by me)



Section north/south through both towers



Section east/west showing the transfer level over the bridge deck



Foundation layout: CFA piles, pile caps and micropile underpinning

A2

BENG DESIGN PROJECT · SONA COLLEGE OF TECHNOLOGY

Steel-intensive archery academy with hostel, Chennai

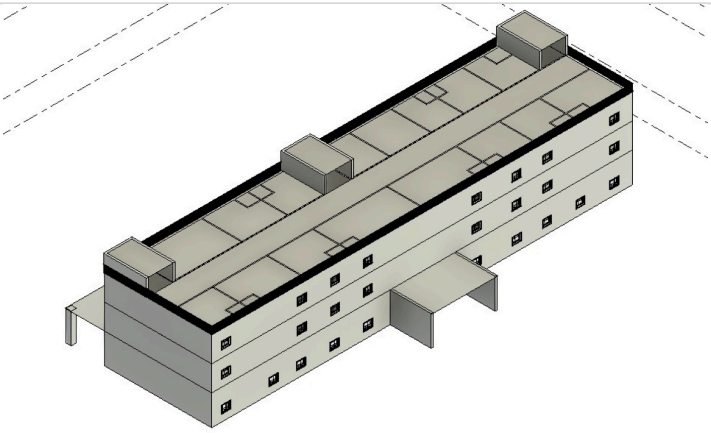
ROLE	LEVEL	CODES	TOOLS
Team of three, planning to estimation	BEng	IS 800, IS 456, IS 875, NBC 2016	STAAD.Pro, STAAD Foundation, Revit, AutoCAD, Excel

A G+2 steel hostel for an archery academy with a 50 m cantilever shooting shed along one side. Columns were only allowed along the long sides, which pushed the design towards long-span steel beams and profiled deck composite floors.

Hostel	50 m × 12.6 m, three storeys
Archery shed	50 m × 7.5 m cantilever, open on three sides
Loads	Dead, live, wind, seismic
Estimate	≈ ₹1.37 crore

WHAT I DID

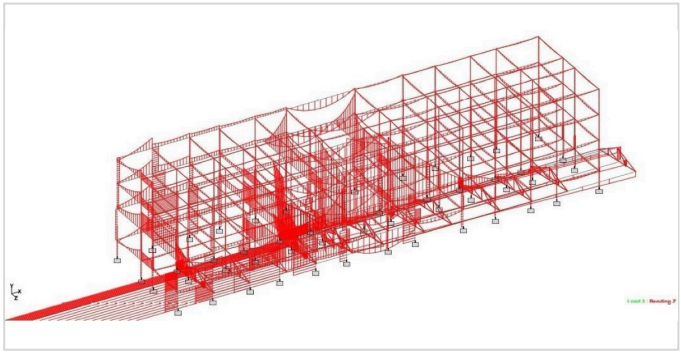
- Drew the plans and sections and set out the column grid.
- Modelled the frame in STAAD.Pro with load combinations to IS 800:2007 and designed the critical beams and columns.
- Checked the critical ISMB 600 beam by hand over 5 m: V = 70 kN against Vd = 944.75 kN, deflection limit L/300 = 16.67 mm.
- Designed the foundations in STAAD Foundation, built the Revit model and prepared the Excel estimate.



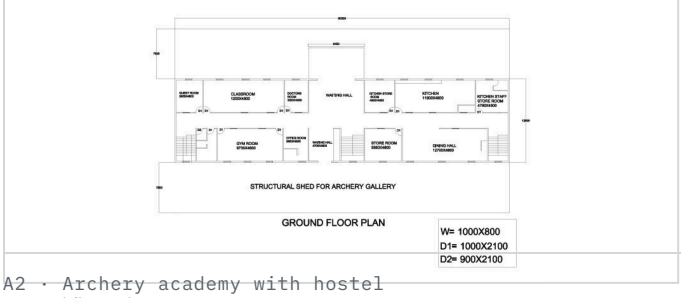
Revit model of the hostel and archery shed



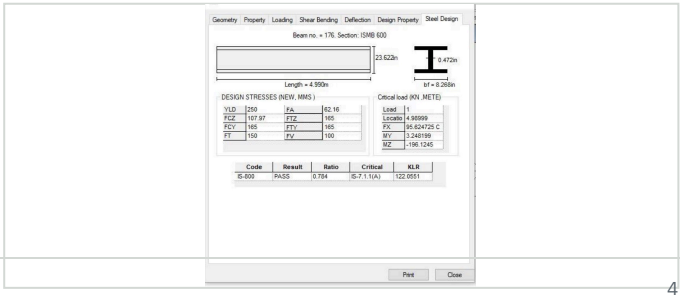
STAAD.Pro steel frame with the cantilever shed



Bending moment diagram



Ground floor plan



ISMB 600 design check to IS 800

B1

MSC DISSERTATION · UNIVERSITY OF LEEDS · SUPERVISOR DR A. RAHMAN

Structural fire performance of WAAM stainless steel RHS columns

ROLE	LEVEL	CODES	TOOLS
Independent research	MSc dissertation	EN 1993-1-4, EN 1993-1-2	Abaqus/CAE 2023, GMNIA, Static Riks

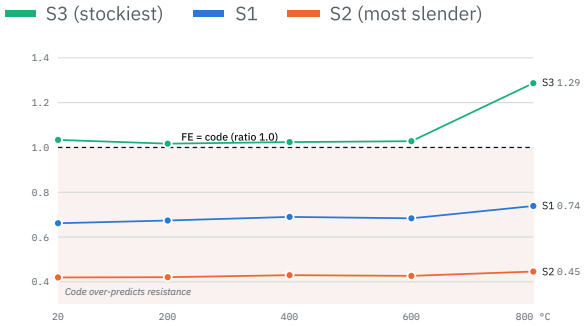
Wire arc additive manufacturing (WAAM) 3D-prints steel members. Its high-temperature material properties are now published, but nobody had checked how WAAM columns buckle in a fire. I asked whether the EN 1993-1-4 column method can be used for them without change.

Sections	S1 150×100×5, S2 120×80×5, S3 200×120×5
Column	2000 mm, pinned at both ends
Temperatures	20, 200, 400, 600, 800 °C
Analyses	15 GMNIA runs, 20 mm S4R mesh
Material	WAAM ER316LSi, Ramberg-Osgood

- Built a two-step GMNIA model: a lateral disturbing force, then Static Riks to peak load.
- Found and fixed a load introduction error during verification. Added stiffened end zones to S3 to stop local crumpling.
- Checked every result against Euler and squash load bounds.

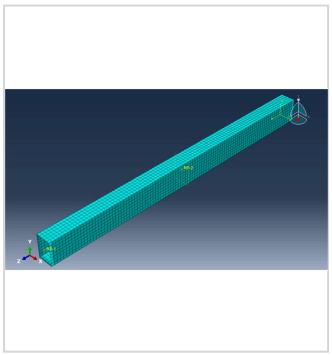
FE resistance as a share of the EN 1993-1-4 prediction

FE/EC3 ratio. Below 1.0 the code over-predicts strength.

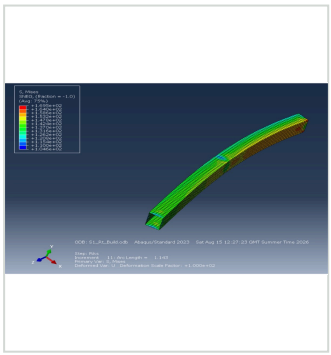


> Slender sections reached as little as 42% of the code prediction while the stockiest met or beat it everywhere. The unmodified EN 1993-1-4 curve cannot yet be called safe for WAAM stainless steel RHS columns in fire.

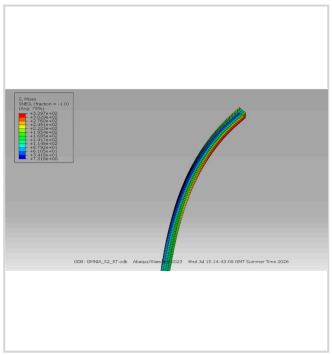
Capacity lost by 800 °C	S1 51.2%, S2 51.5%, S3 46.9%
S3 at 800 °C	421.8 kN against 406.1 kN squash load



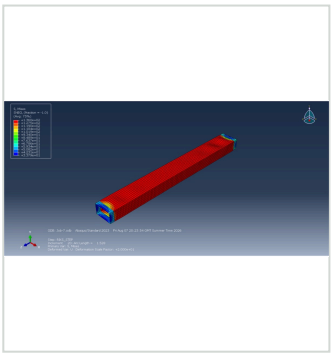
Meshed model, S1



S1 at peak load, 20 °C



S2 at peak load, 20 °C



S3 at 800 °C, squash failure

B2

BENG FINAL YEAR PROJECT · PUBLISHED IN A SCOPUS-INDEXED JOURNAL

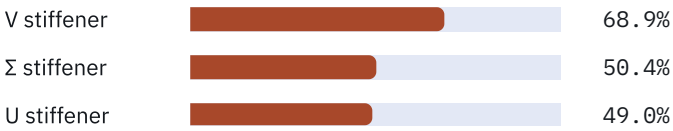
Eccentrically loaded cold-formed steel angle columns

ROLE	LEVEL	METHOD	TOOLS
Team of three, testing and FE modelling	BEng final year	Laboratory tests and FE	ABAQUS (S4R), CUFSM, LVDT

Cold-formed steel angles are light and efficient but sensitive to how the load is applied. We tested angle columns with V, U and Σ shaped intermediate stiffeners under concentric and eccentric compression, then built FE models to extend the results.

- Tested specimens 250 to 1500 mm long at eccentricities up to 120 mm, with an LVDT recording axial shortening.
- Checked local, distortional and global buckling modes in CUFSM.
- Built ABAQUS shell models using eigenvalue buckling then Static Riks.
- Validated against published tests (test/FE 0.96 to 0.98) and our own tests (mean 0.94).

DROP IN BUCKLING LOAD, ECCENTRICITY TOWARDS THE WEAKER AXIS



> Eccentricity towards the stiffer axis changed the capacity only slightly. The study proposed a design curve for practice and was published in July 2025.



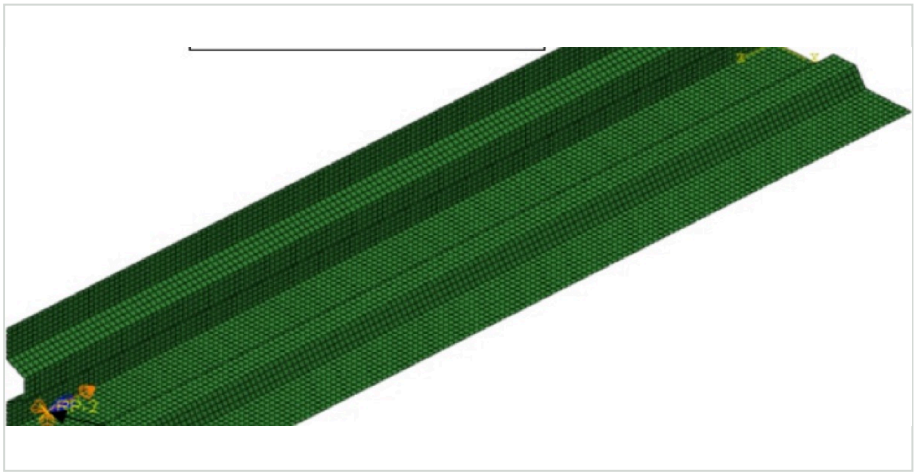
Test set-up with LVDT



Buckled specimen



Σ stiffener specimen after testing



ABAQUS shell model with loading and boundary conditions

B3

BENG PROJECT · NUMERICAL STUDY

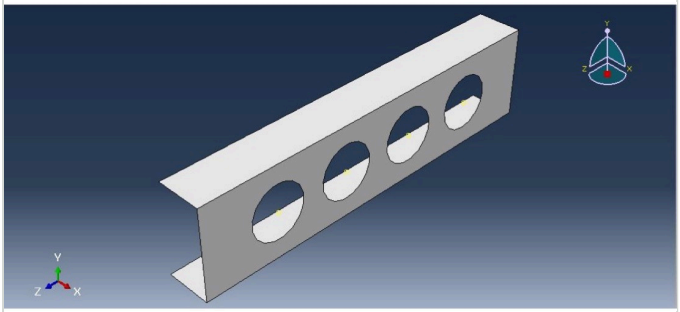
Buckling behaviour of perforated thin-walled columns

ROLE	LEVEL	VARIABLES	TOOLS
Team of three, FE modelling	BEng	Hole shape, opening and spacing ratio	ABAQUS linear and nonlinear

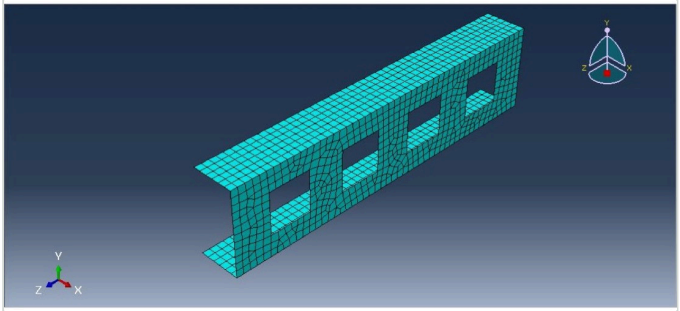
Openings in a column web save weight and let services pass through, but they cut the buckling capacity. We modelled 250 mm long composite C-section columns with circular and square openings to find which layouts keep the most strength.

- Varied hole shape, opening ratio D/D0 and spacing ratio S/D0 against a column without holes.
- Ran linear eigenvalue and nonlinear analyses for buckling and post-buckling.

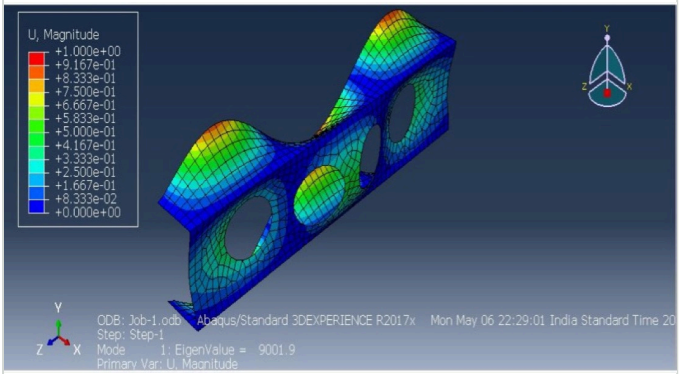
Circular holes	9.6% to 24% loss of buckling load
Square holes	10.2% to 56.4% loss of buckling load
Best circular layout	D/D0 = 1.5, S/D0 = 1.6
Best square layout	D/D0 = 2, S/D0 = 1.67



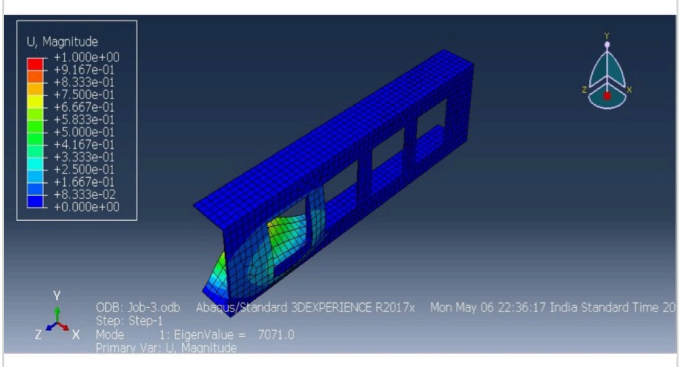
Circular openings



Square openings, meshed



Buckling mode, circular



Buckling mode, square

C Site experience

C1

SITE INTERNSHIP · GOVERNMENT PROJECT

High Court building, Salem

DATE	KIT
Apr – Aug 2025	DGPS, total station, drone

- Carried out site surveys with DGPS, total stations and drones for setting out and progress monitoring.
- Prepared reinforcement detailing for beams, transfer structures and prestressed elements.
- Ran slump and cube tests on around 100 concrete cubes and reported results to the Assistant Engineer.
- Supported full-day concrete pours with quality checks and site safety.
- Wrote daily progress reports and attended progress meetings with contractors.

C2

SITE INTERNSHIP · PUBLIC WORKS DEPARTMENT

Government Law College building, Namakkal

DATE	CONTRACTOR
Jun – Sep 2024	P&C Constructions

- Carried out site and column marking and supervised excavation for isolated footings, checking depths and levels.
- Checked reinforcement for columns, beams and slabs: 16 and 20 mm main bars with 8 mm stirrups at 150 mm.
- Supported shuttering and column concreting placed in layers to avoid segregation.
- Studied the prestressed beams used for spans over 10 m.
- Worked out quantities of cement, aggregates, water and bricks and compared market rates.

D Innovation

PATENT: FRP EMERGENCY SHELTER SYSTEM

A lightweight, modular fibre reinforced polymer shelter that can be put up quickly after a disaster.

SUPERCAPACITOR BRICKS

Fabricated and tested construction bricks with embedded energy storage, looking at how walls could store energy in smart buildings.

· Skills and software

Analysis and design
Hand calculations, ETABS, STAAD.Pro, STAAD Foundation, ABAQUS (GMNIA, Riks, eigenvalue buckling), CUFSM.

Codes
Eurocodes EN 1990 to 1995 and EN 1997 with UK National Annexes, EN 1993-1-2 fire design, IS 800, IS 456, IS 875, NBC 2016.

Drawings and BIM
Revit GA plans, sections and foundation layouts, AutoCAD, SketchUp, Excel estimates.

Sustainability
IStructE Embodied Carbon Calculator, LETI and RIBA 2030 benchmarks, low-carbon material choice.

Site
DGPS, total station and drone surveys, setting out, slump and cube testing, reinforcement checks.

Membership
Graduate Member of the Institution of Structural Engineers, working towards chartership.

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